Analysis of Electronic Electrostatic Generators

Antonio Carlos M. de Queiroz, *Member, IEEE*

***Abstract*—This paper presents a systematic procedure for the formulation of the equations required for the algebraic analysis of electrostatic generators used for energy harvesting purposes. The considered circuits are assumed composed of variable and fixed capacitors, diodes, and switches. Guidelines are presented for the systematic matrix formulation of the required equations, for the calculation of the voltage multiplication factor in the charging transient and for the analysis of the operation at steady state generating energy. Examples are then presented, with the application of the procedure in the analysis of two structures, the first already analyzed by another method, and the last not previously studied in detail.**

Keywords—energy harvesting; electrostatic generators; circuit analysis

I. Introduction

The use of electronic electrostatic generators to extract energy from vibrations has seen some interest in the last years, with devices using several technologies to make the variable capacitors described in the literature, and several structures for the generators proposed [3]-[10]. These generators are composed of one or more variable capacitors, extracting energy from their capacitance variation, and several fixed capacitors and diodes, making an electronic circuit that is unstable when the capacitance variation is large enough. Small starting voltages and charges in the device then grow exponentially along the time, until being large enough for the utilization of the harvested energy, usually by discharging part of the generator using a DC/DC converter. The generator then quickly recovers, starting from the remaining energy, and soon can be discharged again. A basic form of these generators was described in [3], identified as an electronic version of the classical electrostatic device known as "Bennet's doubler" [13]. Several other structures were proposed [7]-[10], aiming at obtaining more energy, working with less capacitance variation, or reducing the voltage in the DC/DC converter. Some were known since long time [14]-[16]. The analysis of the operation of these generators was done in several works, with various degrees of detail and systematic procedure.

In [3], [6], [10], a systematic matrix procedure for the analysis of the generators was presented, using substitution of the diodes by switches, and assuming that the switches only change state between open and closed states at the end of the capacitance variations. The analysis easily computes the ideal, in the absence of losses and electromechanical forces [19], [20], multiplication factor for voltages and charges from cycle to cycle during the startup of the generators, and parameters as the minimum capacitance variation ratio and optimum values for the fixed capacitors. It was shown to work correctly for the electronic Bennet's doubler [3], [2], [3], and for the symmetrical generators in [6]. The method, however, was known to fail when there were certain parasitic capacitances present in the Bennet's doubler circuit, and was verified to not agree correctly with simulations when applied to the "symmetrical unipolar generator", Fig. 1 [9], [10], the first of a family using two-phase voltage multipliers [26].

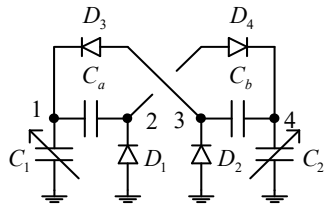


Fig. 1. Symmetrical unipolar variable-capacitance generator, that has a complicated operation during startup.

Another kind of analysis computes the energy that can be extracted from the generator at steady state when it is connected to a DC/DC converter. A typical analysis of this kind is in [17], where the circuit of the electronic Bennet's doubler is considered, as in Fig. 2. The analysis calculated the optimum value for C_1 for any variation of C_a and C_b, by following segmented waveforms produced in the circuit while the diodes conduct or not, in two cases of periodical operation at steady state. The analysis was presented in a relatively informal way. A similar analysis following segments was used for the startup including losses in [18].

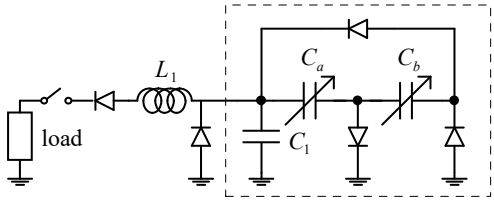


Fig. 2. Electronic Bennet's doubler connected to a buck DC/DC converter.

This work first describes a systematic procedure for the analysis of these electronic electrostatic generators by following segmented waveforms, and then applies it to the analysis of the startup in some cases, aiming specially at the generators described in [6] and [10], the last so far not quantitatively studied in a precise way.

II. Matricial Analysis Using Charge Conservation

A. Startup analysis

Consider first the operation during startup. The generator is considered composed of capacitors and diodes (ideal) only. After an initial transient involving initial charges in the capacitors, all these devices reach a state where all the voltages

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A. C. M. de Queiroz is with the Electrical Engineering Program, COPPE/EP, Federal University of Rio de Janeiro. CP 68504, 21941-972 Rio de Janeiro, RJ, Brazil. (email: acmq@ufrj.br).

and charges grow exponentially, with a fixed multiplication factor z from cycle to cycle. A systematic procedure for the calculation of z may be as follows:

Using a simulation or adequate reasoning, determine the order in which the diodes conduct along a cycle of operation, identifying what happens with the nodal voltages, which become identical when a diode connecting the respective nodes conduct. The result is a series of segments defined by the states of the diodes.

For each segment, write a matrix equation:

$$\mathbf{C}_i\mathbf{e}_i = \mathbf{C}_{i+1}\mathbf{e}_{i+1} \tag{1}$$

Where $\mathbf{C}_i$ and $\mathbf{C}_{i+1}$ are the nodal capacitance matrices during the segment, considering initial, $\mathbf{C}_i$ and final, $\mathbf{C}_{i+1}$, values of the variable capacitances, and $\mathbf{e}_i$, $\mathbf{e}_{i+1}$ are the vectors of nonzero nodal voltages, also initial and final values.

In this system, apply the initial and final conditions that are known, as starting capacitances for the first segment, and equality or annulation of some nodal voltages at the end of the segment. This forms a system of equations that can be solved for the final nodal voltages and final variable capacitances as functions of the initial voltages and capacitances. The system is not always linear, as products of unknown voltages and capacitances may occur.

Repeat the procedure for all the segments, and combine the results to obtain the final nodal voltages, after a complete period, as functions of the initial nodal voltages and of the initial and final capacitances. The expected result is that the nonlinearities shall disappear, and the final voltages at period k, $\mathbf{e}_{k+1}$ are a linear matrix function of the initial voltages $\mathbf{e}_k$:

$$\mathbf{e}_{k+1} = \mathbf{M}\mathbf{e}_k \tag{2}$$

$\mathbf{M}$ is a matrix with coefficients depending on the minimum and maximum values of the capacitances only, and the fixed capacitances. If this result is reached, the desired factor z is just an eigenvalue of $\mathbf{M}$ with value greater than one, because there exists a set of values for $\mathbf{e}_k$, proportional to the eigenvector of $\mathbf{M}$ corresponding to z, so:

$$\mathbf{e}_{k+1} = z\mathbf{e}_k \tag{3}$$

This procedure works with the generators in [3] and [6], at the expense of being somewhat more complicated. It will be shown that it works with the generator in Fig. 1 too, although with some difficulty.

B. *Steady-state analysis*

The other case where the same procedure can be used is in the analysis of the steady-state operation while the generator is connected to a DC/DC converter. The method is the same, requiring first the identification of a particular mode of operation, as the recoveries after a discharge by the converter in one or two cycles used in [17]. As the waveforms repeat at each cycle, there is no need of linearity, and imperfections as diode drop voltages can be included in the analysis (in the startup analysis diode drop voltages would cause the multiplication factor to become dependent on the voltages present on the circuit, complicating the problem). The operation of the converter amounts to the addition of a charge vector $\mathbf{Q}$ to the right side of (1). The final condition is just equality between the initial and final voltage after a complete cycle, $\mathbf{e}_{k+1}$=$\mathbf{e}_k$. The condition leads to a particular value for one capacitor, as C_1 in Fig. 2, so the operation is as desired. The extracted energy can be obtained as function of the equivalent capacitance seen by the DC/DC converter at the end of the cycle, C_{eq}, and of the voltage at which it triggers, v_{lim}, as:

$$E = \frac{1}{2}C_{eq}v_{lim}^{\ 2} \tag{4}$$

The analysis of these cases is usually simpler than the analysis of the startup, but due to lack of space will be detailed in another publication.

III. Startup of Symmetrical Generators

To illustrate the analysis of the startup, consider first the generator derived from another classical device, "Wilson's machine" [14], already analyzed in [6], shown in Fig. 3.

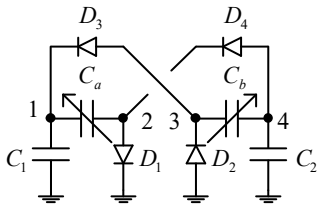


Fig. 3. Symmetrical bipolar generator, derived from "Wilson's machine".

A simulation of the device with C_a and C_b varying complementarily between 60 pF and 360 pF, C_1 = C_2 = 330 pF, and operation at 20 Hz results in the waveforms shown in Fig. 4. As the circuit is symmetrical the segments must be followed just for half period, with the nodal voltages at phase ϕ_2 being related to the nodal voltages at phase ϕ_1 by:

$$\begin{bmatrix} e_{12} \\ e_{22} \end{bmatrix} = -\sqrt{z}\begin{bmatrix} e_{41} \\ e_{11} \end{bmatrix} \tag{5}$$

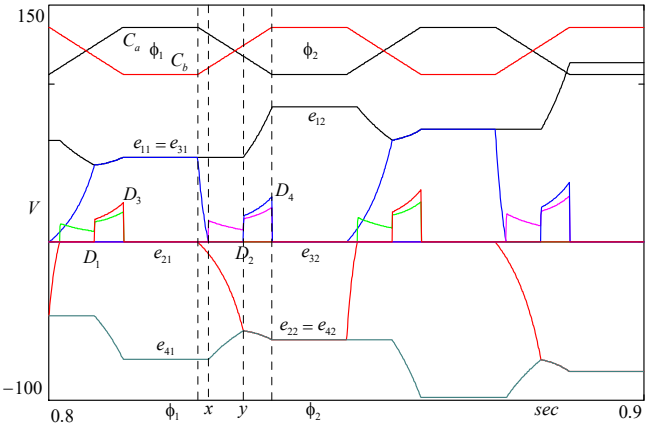


Fig. 4. Simulation of the generator in Fig. 3. The curves show nodal voltages, variable capacitances (100 = 1 nF), currents in the diodes (100 = 0.1 μA), for two cycles at 20 Hz.

For the first segment, $\phi_1 \rightarrow x$, (1) takes the form (6), with the restrictions: $e_{31} = e_{11}$, $e_{21} = 0$, $e_{3x} = 0$, $C_{a1}-C_{ax} = C_{bx}-C_{b1}$, $C_{a1} =$

αC_{min}, $C_{b1} = C_{min}$, and $C_1 = C_2 = \beta C_{min}$.

$$\begin{bmatrix} C_1+C_{a1} & -C_{a1} & 0 & 0 \\ -C_{a1} & C_{a1} & 0 & 0 \\ 0 & 0 & C_{b1} & -C_{b1} \\ 0 & 0 & -C_{b1} & C_2+C_{b1} \end{bmatrix} \begin{bmatrix} e_{11} \\ e_{21} \\ e_{31} \\ e_{41} \end{bmatrix} = \\ = \begin{bmatrix} C_1+C_{ax} & -C_{ax} & 0 & 0 \\ -C_{ax} & C_{ax} & 0 & 0 \\ 0 & 0 & C_{bx} & -C_{bx} \\ 0 & 0 & -C_{bx} & C_2+C_{bx} \end{bmatrix} \begin{bmatrix} e_{1x} \\ e_{2x} \\ e_{3x} \\ e_{4x} \end{bmatrix} \tag{6}$$

For the second segment $x \rightarrow y$, the system is (7), with the additional restrictions $e_{4y} = e_{2y}$ and $C_{a1} - C_{ay} = C_{by} - C_{b1}$.

$$\begin{bmatrix} C_1+C_{ax} & -C_{ax} & 0 \\ -C_{ax} & C_{ax} & 0 \\ 0 & 0 & C_2+C_{bx} \end{bmatrix} \begin{bmatrix} e_{1x} \\ e_{2x} \\ e_{4x} \end{bmatrix} = \\ = \begin{bmatrix} C_1+C_{ay} & -C_{ay} & 0 \\ -C_{ay} & C_{ay} & 0 \\ 0 & 0 & C_2+C_{by} \end{bmatrix} \begin{bmatrix} e_{1y} \\ e_{2y} \\ e_{4y} \end{bmatrix} \tag{7}$$

For the third segment, $y \rightarrow \phi_2$ completing the half cycle, the system is (8), with $C_{a2} = C_{min}$ and $C_{b2} = \alpha C_{min}$.

$$\begin{bmatrix} C_1+C_{ay} & -C_{ay} \\ -C_{ay} & C_{ay}+C_{by}+C_2 \end{bmatrix} \begin{bmatrix} e_{1y} \\ e_{2y} \end{bmatrix} = \\ = \begin{bmatrix} C_1+C_{a2} & -C_{a2} \\ -C_{a2} & C_{a2}+C_{b2}+C_2 \end{bmatrix} \begin{bmatrix} e_{12} \\ e_{22} \end{bmatrix} \tag{8}$$

Solving (6) results in:

$$C_{ax} = C_{min} \frac{e_{11}+\alpha e_{41}}{e_{41}}; e_{1x} = e_{11}; e_{2x} = \frac{e_{11}^{\,2}}{e_{11}+\alpha e_{41}}; e_{4x} = e_{41} \tag{9}$$

Solving (7) with (9) results in $e_{1y} = e_{1x} = e_{11}$, and complicated expressions for C_{ay} and e_{2y}, due to their products:

$$C_{ay} = C_{min} \frac{\left(\begin{array}{l} e_{11}(2\alpha+\beta+2) - e_{41}(\beta+1) - \\ -\sqrt{e_{11}^{\,2}(4\alpha+\beta^2+4\beta+4) - 2e_{11}e_{41}(2\alpha(\beta+1)+\beta^2+3\beta+2) + e_{41}^{\,2}(\beta^2+2\beta+1)} \end{array} \right)}{2e_{11}} \tag{10}$$

$$e_{2y} = \frac{\left(\begin{array}{l} e_{11}\beta + e_{41}(\beta+1) - \\ -\sqrt{e_{11}^{\,2}(4\alpha+\beta^2+4\beta+4) - 2e_{11}e_{41}(\beta+1)(2\alpha+\beta+2) + e_{41}^{\,2}(\beta+1)^2} \end{array} \right)}{2(\alpha+\beta+1)}$$

Solving (8), however, cancels the nonlinear terms, and the final solution reduces to the expected linear form:

$$\begin{bmatrix} e_{12} \\ e_{22} \end{bmatrix} = \frac{1}{\alpha(\beta+1)+\beta(\beta+2)} \begin{bmatrix} \beta+1 & \alpha^2+2\alpha\beta+\beta^2+\beta-1 \\ (\beta+1)^2 & -(\alpha\beta+1) \end{bmatrix} \begin{bmatrix} e_{41} \\ e_{11} \end{bmatrix} \tag{11}$$

The eigenvalues of the matrix **M** are obtained as:

$$\sqrt{z} = \frac{\left(\begin{array}{l} -\beta(\alpha-1) \pm \\ \pm\sqrt{\alpha^2(5\beta^2+8\beta+4) + 2\alpha\beta(4\beta^2+9\beta+6) + \beta^2(4\beta^2+12\beta+9)} \end{array} \right)}{2(\alpha(\beta+1)+\beta(\beta+2))} \tag{12}$$

Substituting numerical values $\alpha = 360/60$ and $\beta = 330/60$:

$$\sqrt{z} = \begin{cases} 0.808875 \\ -1.15155 \end{cases} \tag{13}$$

The negative eigenvalue agrees with the simulation, and with the analysis in [6], where z is calculated exactly as the square of (12). The ratio $e_{41}/e_{11} = e_{12}/e_{22}$ is obtained from the corresponding eigenvector, (14), as −1.38254, also agreeing with the simulation. The analysis in [6] does not calculate it.

$$\frac{e_{41}}{e_{11}} = \frac{\left(\begin{array}{l} -\alpha\beta-\beta-2+ \\ +\sqrt{\alpha^2(5\beta^2+8\beta+4) + 2\alpha\beta(4\beta^2+9\beta+6) + \beta^2(4\beta^2+12\beta+9)} \end{array} \right)}{2(\beta+1)^2} \tag{14}$$

Consider now the symmetrical unipolar generator in Fig. 1. A simulation of its startup, for C_1 and C_2 varying complementarily between 60 pF and 360 pF, $C_a = C_b = 330$ pF, and operation at 20 Hz results in the waveforms shown in Fig. 5. All the voltages are now positive.

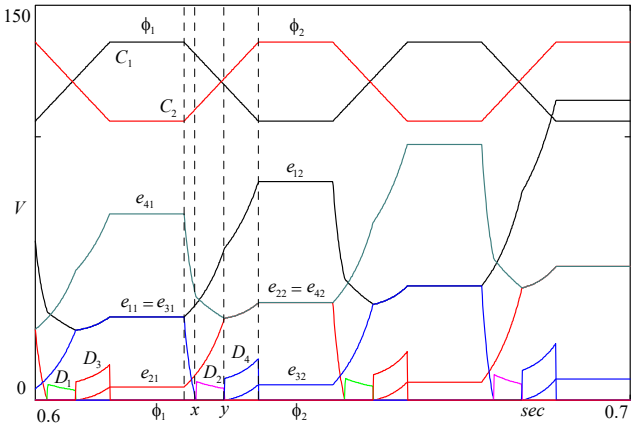


Fig. 5. Simulation of the unipolar generator in Fig. 1, as in Fig. 4.

The symmetry can be used again, with just half of the cycle followed. There are also three segments to follow, with two complications: First, there are three nonzero voltages instead of two to consider, and second, all the diodes conduct separately, without overlap. The relation to be obtained, for a certain ratio of the three voltages, is:

$$\begin{bmatrix} e_{22} \\ e_{32} \\ e_{12} \end{bmatrix} = \sqrt{z} \begin{bmatrix} e_{11} \\ e_{21} \\ e_{41} \end{bmatrix} \tag{15}$$

The analysis is similar to the previous case for the first two segments, with C_1 and C_2 varying instead of C_a and C_b, which are fixed in this case. In the first segment, $\phi_1 \rightarrow x$, (1) takes the form (16), with the restrictions: $e_{31} = e_{11}$, $e_{3x} = 0$, $C_{11} - C_{1x} = C_{2x} - C_{21}$, $C_{11} = \alpha C_{min}$, $C_{21} = C_{min}$, and $C_a = C_b = \beta C_{min}$.

$$\begin{bmatrix} C_{11}+C_a & -C_a & 0 & 0 \\ -C_a & C_a & 0 & 0 \\ 0 & 0 & C_b & -C_b \\ 0 & 0 & -C_b & C_{21}+C_b \end{bmatrix} \begin{bmatrix} e_{11} \\ e_{21} \\ e_{31} \\ e_{41} \end{bmatrix} = \\ = \begin{bmatrix} C_{1x}+C_a & -C_a & 0 & 0 \\ -C_a & C_a & 0 & 0 \\ 0 & 0 & C_b & -C_b \\ 0 & 0 & -C_b & C_{2x}+C_b \end{bmatrix} \begin{bmatrix} e_{1x} \\ e_{2x} \\ e_{3x} \\ e_{4x} \end{bmatrix} \tag{16}$$

For the second segment $x \rightarrow y$, the system is (17), with the restrictions $e_{4y} = e_{2y}$ and $C_{11} - C_{1y} = C_{2y} - C_{21}$. For the third segment, $y \rightarrow \phi_2$, there is a difference, as D_1 does not conduct. The system is (18), with $e_{3y} = 0$, $C_{12} = C_{min}$ and $C_{22} = \alpha C_{min}$.

$$\begin{bmatrix} C_{1x}+C_a & -C_a & 0 \\ -C_a & C_a & 0 \\ 0 & 0 & C_{2x}+C_b \end{bmatrix}\begin{bmatrix} e_{1x} \\ e_{2x} \\ e_{4x} \end{bmatrix} = \\ = \begin{bmatrix} C_{1y}+C_a & -C_a & 0 \\ -C_a & C_a & 0 \\ 0 & 0 & C_{2y}+C_b \end{bmatrix}\begin{bmatrix} e_{1y} \\ e_{2y} \\ e_{4y} \end{bmatrix} \tag{17}$$

$$\begin{bmatrix} C_{1y}+C_a & -C_a & 0 \\ -C_a & C_{2y}+C_a+C_b & -C_b \\ 0 & -C_b & C_b \end{bmatrix}\begin{bmatrix} e_{1y} \\ e_{2y} \\ e_{3y} \end{bmatrix} = \\ = \begin{bmatrix} C_{12}+C_a & -C_a & 0 \\ -C_a & C_{22}+C_a+C_b & -C_b \\ 0 & -C_b & C_b \end{bmatrix}\begin{bmatrix} e_{12} \\ e_{22} \\ e_{32} \end{bmatrix} \tag{18}$$

Solving (16):

$$C_{2x} = C_{min}\frac{e_{41}}{e_{41}-e_{11}};\; e_{1x} = \frac{e_{11}\alpha(e_{11}-e_{41})}{e_{11}(\alpha+1)-e_{41}\alpha}; \\ e_{2x} = \frac{{e_{11}}^2+e_{11}e_{21}(\alpha+1)+e_{21}e_{41}\alpha}{e_{41}\alpha-e_{11}(\alpha+1)};\; e_{4x} = e_{41}-e_{11} \tag{19}$$

The solution of (17) using (19) results in (20), presenting a problem similar to the previous case, due to the products of unknown capacitances and voltages. To simplify the expressions a bit, let $\gamma = \beta+1$ and $\lambda = \alpha+\beta+1$.

$$C_{2y} = C_{min}\frac{\left(e_{11}-e_{21}(\alpha-\beta+1)-e_{41}\gamma+\right.}{2(e_{11}-e_{21})} \\ \frac{\left.+\sqrt{{e_{11}}^2(1-4\alpha\beta)+2e_{11}(e_{41}\gamma(2\alpha+1)-e_{21}\lambda)+(e_{21}\lambda-e_{41}\gamma)^2}\right)}{2(e_{11}-e_{21})} \\ e_{1y} = \frac{\left(e_{11}(2\alpha+1)-e_{21}\lambda-e_{41}\gamma+\right.}{2\lambda} \\ \frac{\left.+\sqrt{{e_{11}}^2(1-4\alpha\beta)+2e_{11}(e_{41}\gamma(2\alpha+1)-e_{21}\lambda)+(e_{21}\lambda-e_{41}\gamma)^2}\right)}{2\lambda} \\ e_{2y} = \frac{\left(-e_{11}(2\beta+1)+e_{21}\lambda+e_{41}\gamma+\right.}{2\lambda} \\ \frac{\left.+\sqrt{{e_{11}}^2(1-4\alpha\beta)+2e_{11}(e_{41}\gamma(2\alpha+1)-e_{21}\lambda)+(e_{21}\lambda-e_{41}\gamma)^2}\right)}{2\lambda} \tag{20}$$

In this case the solution for the last segment, (18) using (20), does not eliminate the nonlinear terms, resulting in (21), so a solution in the form (2) does not exist, due to the irreducible square roots. However, the equations (21) also result in $\mathbf{e}_2$ proportional to $\mathbf{e}_1$ for a certain ratio of the voltages, even with the square roots. It is known from the simulations and experiments, and from the linearity of the elements in the model, that a solution in the forms (3) or (15) must exist. It must be found numerically in this case.

$$e_{12} = \frac{-\left(\begin{array}{l}-e_{11}(2\alpha^3+2\alpha^2(3\beta+1)+2\alpha\beta(\beta+2)-\beta^2)+\beta(e_{21}(2\alpha^2+\alpha(3\beta+2)+\beta\gamma)-e_{41}\gamma(2\alpha+\beta+2))+ \\ +\beta^2\sqrt{{e_{11}}^2(1-4\alpha\beta)+2e_{11}(e_{41}\gamma(2\alpha+1)-e_{21}\lambda)+{e_{21}}^2(\alpha^2+2\alpha\gamma+\gamma^2)-2e_{21}e_{41}\lambda\gamma+{e_{41}}^2\gamma^2}\end{array}\right)}{2\lambda(\alpha\gamma+\beta)} \\ e_{22} = \frac{-\left(\begin{array}{l}-e_{11}\beta(2\alpha^2-2\alpha-3\gamma)+e_{21}\beta\lambda(\beta-1)-e_{41}\gamma^2(2\alpha+\beta+2))+ \\ +\beta\gamma\sqrt{{e_{11}}^2(1-4\alpha\beta)+2e_{11}(e_{41}\gamma(2\alpha+1)-e_{21}\lambda)+{e_{21}}^2(\alpha^2+2\alpha\gamma+\gamma^2)-2e_{21}e_{41}\lambda\gamma+{e_{41}}^2\gamma^2}\end{array}\right)}{2\lambda(\alpha\gamma+\beta)} \\ e_{32} = \frac{-\left(\begin{array}{l}-e_{11}(2\alpha^2\beta+\alpha(2\beta^2+\beta+1)-\beta(\beta+2))+e_{21}(\alpha^2\gamma+\alpha(2\beta^2+2\beta+1)+\beta^2\gamma)-e_{41}\gamma(\alpha\gamma+\beta^2+2\beta+2) \\ +(\alpha\gamma+\beta(\beta+2))\sqrt{{e_{11}}^2(1-4\alpha\beta)+2e_{11}(e_{41}\gamma(2\alpha+1)-e_{21}\lambda)+{e_{21}}^2(\alpha^2+2\alpha\gamma+\gamma^2)-2e_{21}e_{41}\lambda\gamma+{e_{41}}^2\gamma^2}\end{array}\right)}{2\lambda(\alpha\gamma+\beta)} \tag{21}$$

Let the three variables be in the proportion $e_{11}:e_{21}:e_{41} = 1:a:b$, forming a kind of generalized normalized eigenvalue for the functions (21). A system of equations to solve for z, a, and b is, where $\mathbf{F}(\mathbf{e})$ are the functions (21), is:

$$\mathbf{F}\left(\begin{bmatrix} 1 \\ a \\ b \end{bmatrix}\right) - \sqrt{z}\begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{22}$$

This nonlinear system, in the form $\mathbf{G}(\mathbf{x}) = \mathbf{0}$, is readily solved numerically by the Newton-Raphson method, where $\mathbf{J}(\mathbf{G})$ is the jacobian matrix of $\mathbf{G}$, and $\mathbf{x}$ is the vector $[\sqrt{z}\ a\ b]^T$:

$$\mathbf{J}(\mathbf{G}(\mathbf{x}_n))\mathbf{x}_{n+1} = -\mathbf{G}(\mathbf{x}_n) + \mathbf{J}(\mathbf{G}(\mathbf{x}_n))\mathbf{x}_n \tag{23}$$

The method solves a system of linear equations to find $\mathbf{x}_{n+1}$, departing from an approximation $\mathbf{x}_n$. Iterating the solution a few times, with $\alpha = 360/60$ and $\beta = 330/60$, starting from $\mathbf{x} = [1\ 2\ 3]^T$, the solutions agree perfectly with the simulations:

$$a = \frac{e_{21}}{e_{11}} = 0.159674;\quad b = \frac{e_{41}}{e_{11}} = 2.24117;\quad \sqrt{z} = 1.17138$$

Fig. 6 shows an experimental measurement of the structure in Fig.1, using the generator described in [6], with e_1 and e_4 measured through 82 pF – 470 nF capacitive dividers and 10 MΩ oscilloscope probes. C_{min} is then ~60+82 pF. The predicted z is equal to 1.1538. The curves show ~1.08, smaller due to unaccounted losses and distributed capacitances.

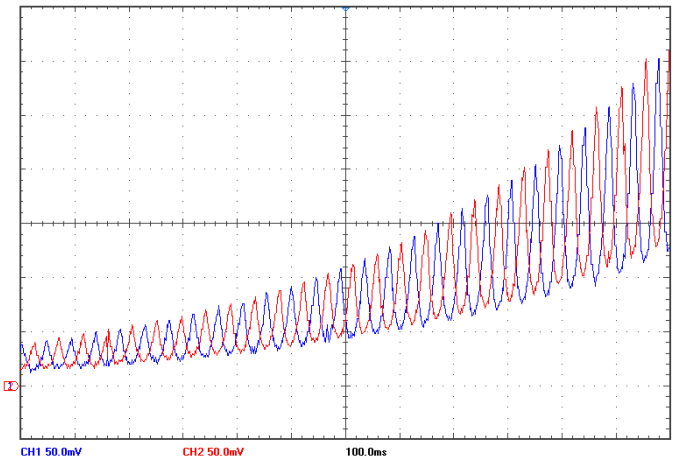


Fig. 6. Experimental measurement of the startup for the structure in Fig. 1.

IV. Conclusions

A systematic method for the formulation of charge-conservation equations for the analysis of electronic electrostatic generators was presented. It can calculate how the generator starts in the ideal case of no losses as a kind of eigenvalues and eigenvectors problem, and also how it operates in steady state generating energy. The method was shown to work correctly even when some diodes do not conduct at the ends of the capacitance variations. The latter case usually results in nonlinear equations, requiring numerical solution. The analysis confirmed the observation that the structure of the "unipolar generator" results in the fastest startup among the structures using two variable capacitors described so far, confirming the promising results in energy harvesting from this kind of generator observed in [9], [10].

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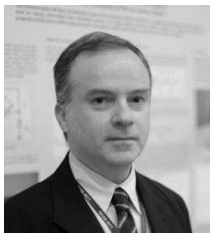
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Antonio Carlos M. de Queiroz (M'79) received the B. Sc. EE. Degree in 1979, the M. Sc. EE. Degree in 1984, and the D. Sc. EE. Degree in 1990 from the Universidade Federal do Rio de Janeiro, Rio de Janeiro, Brazil.

Since 1982 he has been with the same institution, where he is now Titular Professor at the Department of Electronic Engineering and at the Electrical Engineering Program of the Coordenação dos Programas de Pós-graduação em Engenharia (COPPE). His main interests are in general circuit theory, filters and signal processing, computer-aided synthesis and analysis of circuits, high-voltage devices, energy harvesting using electrostatic generators, and the history of electricity.